

 Yildiz Technical University Faculty of Chemical and Metallurgical Engineering Department of Mathematical Engineering Midterm Exam Question and Solution Sheet	SCORE TABLE						
	1	2	3	SUM			
Name-Surname							
Number							
Department							Date
							23.11.2023
Lecture	MTM1531 LINEAR ALGEBRA	Grup Number		Duration	100 min.	Room	
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You must show all details of your work to get full credit. Correct answers with no detail will get zero credit.							

1- (30 p) Let $W = \left\{ \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix} : 2a+b+d=c \right\} \subseteq M_{3 \times 2}(\mathbb{R})$.

A. Is subset W a subspace of $M_{3 \times 2}(\mathbb{R})$?

B. Find a basis and dimension of W . (Satisfy linear independent and span axioms)

A. i) Let $a=b=c=d$ and $2 \cdot 0 + 0 + 0 = 0$, so $\begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \in W \neq \emptyset$ (3)

ii) Let $w_1 = \begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \\ e_1 & f_1 \end{bmatrix}, w_2 = \begin{bmatrix} a_2 & b_2 \\ c_2 & d_2 \\ e_2 & f_2 \end{bmatrix} \in W$ such that $2a_1+b_1+d_1=c_1$ and $2a_2+b_2+d_2=c_2$

$w_1 + w_2 = \begin{bmatrix} a_1+a_2 & b_1+b_2 \\ c_1+c_2 & d_1+d_2 \\ e_1+e_2 & f_1+f_2 \end{bmatrix}$ and since $(2a_1+b_1+d_1) + (2a_2+b_2+d_2) = c_1+c_2$, so $w_1 + w_2 \in W$ (6)

iii) Let k be any scalar and $w_1 \in W$ s.t. $2a_1+b_1+d_1=c_1$

$k \cdot w_1 = \begin{bmatrix} ka_1 & kb_1 \\ kc_1 & kd_1 \\ ke_1 & kf_1 \end{bmatrix}$ and since $k(2a_1+b_1+d_1) = k \cdot c_1$, so $k \cdot w_1 \in W$ (6)

For this reason, W is a subspace of $M_{3 \times 2}(\mathbb{R})$.

$$3. \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix} = \begin{bmatrix} a & b \\ 2a+b & d \\ e & f \end{bmatrix} = a \cdot \underbrace{\begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 0 & 0 \end{bmatrix}}_{v_1} + b \cdot \underbrace{\begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 0 \end{bmatrix}}_{v_2} + d \cdot \underbrace{\begin{bmatrix} 0 & 0 \\ 1 & 1 \\ 0 & 0 \end{bmatrix}}_{v_3} + e \cdot \underbrace{\begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \end{bmatrix}}_{v_4} + f \cdot \underbrace{\begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}}_{v_5}$$

$$\text{span} \{v_1, v_2, v_3, v_4, v_5\} = W$$

$$a_1 v_1 + a_2 v_2 + a_3 v_3 + a_4 v_4 + a_5 v_5 = 0$$

$$a_1 \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 0 & 0 \end{bmatrix} + a_2 \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 0 \end{bmatrix} + a_3 \begin{bmatrix} 0 & 0 \\ 1 & 1 \\ 0 & 0 \end{bmatrix} + a_4 \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \end{bmatrix} + a_5 \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a_1 & a_2 \\ 2a_1 + a_2 + a_3 & a_3 \\ a_4 & a_5 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \Rightarrow \begin{cases} a_1 = 0, a_2 = 0, a_4 = 0, a_5 = 0 \\ 2a_1 + a_2 + a_3 = 0, a_3 = 0 \end{cases}$$

We found $a_1 = a_2 = a_3 = a_4 = a_5 = 0$, so $\{v_1, v_2, v_3, v_4, v_5\}$ is a linearly independent.

Since $\mathcal{B} = \{v_1, v_2, v_3, v_4, v_5\}$ satisfy span and linearly independent axioms, we can say that $\mathcal{B} = \{v_1, v_2, v_3, v_4, v_5\}$ is a basis for W subspace.

$$\dim W = 5$$

$$c_1 v_1 + c_2 v_2 + c_3 v_3 + c_4 v_4 + c_5 v_5 = v$$

$$c_1 \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 0 & 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 0 \end{bmatrix} + c_3 \begin{bmatrix} 0 & 0 \\ 1 & 1 \\ 0 & 0 \end{bmatrix} + c_4 \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \end{bmatrix} + c_5 \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} c_1 & c_2 \\ 2c_1 + c_2 + c_3 & c_3 \\ c_4 & c_5 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix} \Rightarrow \begin{cases} c_1 = a \\ c_2 = b \\ c_3 = d \\ c_4 = e \\ c_5 = f \end{cases}$$

For this reason, $\text{span} \{v_1, v_2, v_3, v_4, v_5\} = W$.

2- A. (20 pt) Let $v_1 = t^2 + t + 2$, $v_2 = 2t^2 + t$ and $v_3 = 3t^2 + 2t + 2$ on vector space P_2 , these vectors are linearly independent? Explain with your reasons?

B. (15 pt) Find the inverse of $A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 4 & -1 & 3 \end{bmatrix}$ by using adjoint matrix. ($A^{-1} = ?$)

A. To show that $\{v_1, v_2, v_3\}$ is a linearly independent, satisfy

$$a_1 v_1 + a_2 v_2 + a_3 v_3 = 0 \quad (1)$$

$$a_1(t^2 + t + 2) + a_2(2t^2 + t) + a_3(3t^2 + 2t + 2) = 0t^2 + 0t + 0 \quad (2)$$

$$t^2(a_1 + 2a_2 + 3a_3) + t(a_1 + a_2 + 2a_3) + (2a_1 + 2a_3) = 0t^2 + 0t + 0 \quad (2)$$

$$\begin{cases} a_1 + 2a_2 + 3a_3 = 0 \\ a_1 + a_2 + 2a_3 = 0 \\ 2a_1 + 2a_3 = 0 \end{cases} \quad \text{homogeneous linear equation system} \quad (1)$$

$$|A| = \begin{vmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 4 & -1 & 3 \end{vmatrix} = 0, \text{ since } |A| = 0, \text{ then } \{v_1, v_2, v_3\} \text{ vector system is a linearly dependent.} \quad (2) \quad (1)$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 3 & 1 & 2 & 0 \\ 4 & -1 & 3 & 0 \end{array} \right] \xrightarrow[R_{32}(-2)]{R_{21}(-1)} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & -1 & -1 & 0 \\ 0 & -2 & -2 & 0 \end{array} \right] \xrightarrow[R_{32}(2)]{R_2(-1)} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \xrightarrow{R_{12}(-2)} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \quad (5)$$

$$r_A = r_{[A:B]} = 2 < 3 = n$$

$$n - r = 3 - 2 = 1 \text{ parameter} \quad (2)$$

$$a_1 + a_3 = 0 \Rightarrow a_1 = -a_3 \quad (2)$$

$$a_2 + a_3 = 0 \Rightarrow a_2 = -a_3$$

$$\text{General solution } (a_1, a_2, a_3) = (-a_3, -a_3, a_3)$$

$$\text{Specifically } a_3 = -1 \Rightarrow a_1 = 1, a_2 = 1 \text{ and } (1, 1, -1) \quad (1)$$

Then $\{v_1, v_2, v_3\}$ is a linearly dependent

$$a_1 v_1 + a_2 v_2 + a_3 v_3 = 0$$

$$1v_1 + 1v_2 - 1v_3 = 0 \Rightarrow v_1 + v_2 = v_3 \quad (1)$$

8. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 4 & -1 & 3 \end{bmatrix}$

$$c_{11} = (-1)^{1+1} \begin{vmatrix} 1 & 2 \\ -1 & 3 \end{vmatrix} = 1 \cdot (3 - (-2)) = 5 \quad c_{21} = (-1)^{2+1} \begin{vmatrix} 2 & 3 \\ -1 & 3 \end{vmatrix} = -9 \quad c_{31} = (-1)^{3+1} \begin{vmatrix} 2 & 3 \\ 1 & 2 \end{vmatrix} = 1$$

$$c_{12} = (-1)^{1+2} \begin{vmatrix} 3 & 2 \\ 4 & 3 \end{vmatrix} = -1 \quad c_{22} = (-1)^{2+2} \begin{vmatrix} 1 & 3 \\ 4 & 3 \end{vmatrix} = -9 \quad c_{32} = (-1)^{3+2} \begin{vmatrix} 1 & 3 \\ 3 & 2 \end{vmatrix} = 7$$

$$c_{13} = (-1)^{1+3} \begin{vmatrix} 3 & 1 \\ 4 & -1 \end{vmatrix} = -7 \quad c_{23} = (-1)^{2+3} \begin{vmatrix} 1 & 2 \\ 4 & -1 \end{vmatrix} = 9 \quad c_{33} = (-1)^{3+3} \begin{vmatrix} 1 & 2 \\ 3 & 1 \end{vmatrix} = -5$$

$$\text{adj}(A) = \begin{bmatrix} 5 & -1 & -7 \\ -9 & -9 & 9 \\ 1 & 7 & -5 \end{bmatrix}^T = \begin{bmatrix} 5 & -9 & 1 \\ -1 & -9 & 7 \\ -7 & 9 & -5 \end{bmatrix} \quad (1)$$

$$|A| = \begin{vmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 4 & -1 & 3 \end{vmatrix} = (-1)^{1+1} \begin{vmatrix} 1 & 2 \\ -1 & 3 \end{vmatrix} + 2 \cdot (-1)^{1+2} \begin{vmatrix} 3 & 2 \\ 4 & 3 \end{vmatrix} + 3 \cdot (-1)^{1+3} \begin{vmatrix} 3 & 1 \\ 4 & -1 \end{vmatrix} = -18 \quad (3)$$

$$A^{-1} = \frac{\text{adj}(A)}{|A|} = \frac{-1}{18} \begin{bmatrix} 5 & -9 & 1 \\ -1 & -9 & 7 \\ -7 & 9 & -5 \end{bmatrix} = \begin{bmatrix} -5/18 & 1/2 & -1/18 \\ 1/18 & 1/2 & -7/18 \\ 7/18 & -1/2 & 5/18 \end{bmatrix} \quad (2)$$

3- A. (25 pt) Let $E = \{(1, 2, -1, 6), (3, -1, 2, 11), (2, 5, -4, -20)\} \subseteq \mathbb{R}^4$.

- Find a basis for row space of spanned by E of $\mathbb{R}^4$ and find rank.
- Find a basis for column space of spanned by E of $\mathbb{R}^4$ and find rank. (Write the columns as rows and find the base.)

B. (10 pt) Let $v = (3, -4, 5)$ and $w = (1, 2, -2)$ on $\mathbb{R}^3$. Find the angle and distance between these vectors.

A. Since elementary row operations do not change the row space of matrix, we can find a basis for the row space of $\bar{E}$. Reducing $\bar{E}$ to row-echelon form, we obtain

$$\bar{E} = \begin{bmatrix} 1 & 2 & -1 & 6 \\ 3 & -1 & 2 & 11 \\ 2 & 5 & -4 & -20 \end{bmatrix} \xrightarrow{\substack{R_{21}(-3) \\ R_{31}(-2)}} \begin{bmatrix} 1 & 2 & -1 & 6 \\ 0 & -7 & 5 & -7 \\ 0 & 1 & -2 & -32 \end{bmatrix} \xrightarrow{R_{23}} \begin{bmatrix} 1 & 2 & -1 & 6 \\ 0 & 1 & -2 & -32 \\ 0 & -7 & 5 & -7 \end{bmatrix} \xrightarrow{\substack{R_{12}(-2) \\ R_{32}(7)}} \begin{bmatrix} 1 & 0 & 3 & 70 \\ 0 & 1 & -2 & -32 \\ 0 & 0 & -9 & -231 \end{bmatrix}$$

$$\xrightarrow{R_3(-1/9)} \begin{bmatrix} 1 & 0 & 3 & 70 \\ 0 & 1 & -2 & -32 \\ 0 & 0 & 1 & 231/9 \end{bmatrix} \quad (6)$$

The nonzero row vectors a basis for the row space and hence form a basis for the row space

$$\begin{aligned} r_1 &= [1 \ 0 \ 3 \ 70] \\ r_2 &= [0 \ 1 \ -2 \ -32] \\ r_3 &= [0 \ 0 \ 1 \ 231/9] \end{aligned} \quad \left. \begin{array}{l} \{r_1, r_2, r_3\} \text{ is a basis of row space of spanned by } \bar{E} \text{ of } \mathbb{R}^4 \\ \text{rank } \bar{E} = 3 \text{ and } \dim \bar{E} = 3 \end{array} \right\} \quad (2) \quad (1)$$

ii) To find column space, take transpose of E .

$$\begin{aligned}
 E^T &= \begin{bmatrix} 1 & 3 & 2 \\ 2 & -1 & 5 \\ -1 & 2 & -4 \\ 6 & 11 & -20 \end{bmatrix} \xrightarrow{\substack{R_{21}(-2) \\ R_{31}(1) \\ R_{41}(-6)}} \begin{bmatrix} 1 & 3 & 2 \\ 0 & -7 & 1 \\ 0 & 5 & -2 \\ 0 & -7 & -32 \end{bmatrix} \xrightarrow{R_2(-1/7)} \begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & -1/7 \\ 0 & 5 & -2 \\ 0 & -7 & -32 \end{bmatrix} \\
 &\xrightarrow{\substack{R_1(-5) \\ R_{42}(7)}} \begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & -1/7 \\ 0 & 0 & -9/7 \\ 0 & 0 & -33 \end{bmatrix} \xrightarrow{R_3(-7/9)} \begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & -1/7 \\ 0 & 0 & 1 \\ 0 & 0 & -33 \end{bmatrix} \xrightarrow{R_{43}(33)} \begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & -1/7 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \quad (7)
 \end{aligned}$$

$$\begin{aligned}
 w_1 &= [1 \ 3 \ 2]^T \\
 w_2 &= [0 \ 1 \ -1/7]^T \\
 w_3 &= [0 \ 0 \ 1]^T
 \end{aligned}
 \left. \vphantom{\begin{aligned} w_1 \\ w_2 \\ w_3 \end{aligned}} \right\} (3) \{w_1, w_2, w_3\} \text{ is a basis of column space.}$$

$\text{rank } E^T = 3$ [rank of column space] (2)
 $\dim V = 3$ [dimension of column space] (1)

As a result, we see that rank of row space is equal to rank of column space.

8. ✓ $v \cdot w = |v| \cdot |w| \cdot \cos \theta$

$$\begin{aligned}
 v \cdot w &= 3 \cdot 1 + (-4) \cdot 2 + 5 \cdot 2 = -15 \\
 |v| &= \sqrt{3^2 + (-4)^2 + 5^2} = 5\sqrt{2} \\
 |w| &= \sqrt{1^2 + 2^2 + 2^2} = \sqrt{9} = 3 \\
 \cos \theta &= \frac{v \cdot w}{|v| \cdot |w|} = \frac{-15}{5\sqrt{2} \cdot 3} = \frac{-1}{\sqrt{2}} = -\frac{\sqrt{2}}{2}
 \end{aligned}
 \quad (7)$$

$$\Rightarrow \arccos\left(-\frac{\sqrt{2}}{2}\right) = \frac{3\pi}{4}$$

✓

$$\begin{aligned}
 d(v, w) &= \sqrt{(3-1)^2 + (-4-2)^2 + (5-2)^2} \\
 d(v, w) &= \sqrt{2^2 + (-6)^2 + 3^2} \\
 d(v, w) &= \sqrt{4 + 36 + 9} \\
 d(v, w) &= \sqrt{49}
 \end{aligned}
 \quad (3)$$